

ACADEMIC ESSAY

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Building a Crisis

How Dhaka's construction boom became an uncounted source of the air its residents breathe

NSC2283: Environmental Science

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Portfolio note

This essay grew out of a group video project for Environmental Science, for which I shot, scripted, edited, and arranged the expert interview of a news-style package on construction-driven air pollution in Dhaka. I have rebuilt the accompanying brief here as an argument-driven academic essay rather than a set of production notes. It draws on the reporting the package was built from alongside peer-reviewed and monitoring sources. Every figure has been checked against a named source and attributed; where estimates differ across sources, the range is reported rather than a single convenient number. The finished package is linked in the references and on the portfolio page.

Abstract

Public debate about Dhaka's notoriously polluted air tends to fixate on a single visible culprit, the brick kilns ringing the city, while a source that is literally everywhere goes unnamed: the construction boom rebuilding the city itself. This essay argues that construction is a major, seasonally amplified, and systematically under-acknowledged contributor to Dhaka's fine-particulate burden, and that its invisibility is not an accident of perception but a predictable consequence of how pollution is attributed and regulated. It sets out the science of particulate matter and its health effects, locates construction and road dust within the established source apportionment of Dhaka's PM_{2.5}, explains why the delta's geology and the dry-season meteorology compound the problem, and considers why a source this large remains politically and perceptually cheap to ignore. It closes by asking what it would take to make construction dust count — and reflects on why a news-style video package, rather than a lecture, was the right instrument for making a form of harm people walk through every day visible to them again.

1. The pollution people walk through

Dhaka's air is among the worst of any major city on earth, and the public conversation about why tends to converge on the brick kilns that ring the city in the dry season. The kilns are a real and serious source, but their prominence in the public mind has a side effect: it crowds out sources that are less photogenic and less seasonal in the popular imagination, even when they are neither less harmful nor less present. The most important of these is the construction boom itself. Dhaka is being rebuilt more or less continuously — residential towers, the metro rail, the elevated expressway, the endless smaller sites that fill in between them — and the dust thrown up by that rebuilding, the broken roads around every site, the demolition debris, and the uncovered trucks of sand and aggregate are a daily, street-level contribution to what residents breathe.

The difficulty is precisely that this pollution is ordinary. Dust is the kind of harm people walk through without registering it as harm; it looks like a nuisance, or like the unavoidable texture of a growing city, rather than a health exposure. This essay takes the position that construction is a first-order air-pollution problem for Dhaka, and that the reason it is treated as a second-order one has more to do with how the problem is seen, counted, and assigned than with the physics of the particles involved.

2. What the particles are, and why they matter

The scientific picture underlying the concern is settled. The pollutant of greatest health significance is fine particulate matter, PM_{2.5} — particles 2.5 micrometres in diameter or smaller, fine enough to lodge deep in the lungs and to cross from the alveoli into the bloodstream. Because they bypass the body's coarse-filtering defences, these particles are causally linked to stroke, ischaemic heart disease, chronic obstructive pulmonary disease, lung cancer, and lower-respiratory infection; the World Health Organization attributes millions of premature deaths worldwide each year to ambient air pollution of this kind (World Health Organization, 2021).

Against that backdrop Dhaka's numbers are stark. The World Health Organization's 2021 guideline sets the safe annual-average PM_{2.5} concentration at 5 micrograms per cubic metre. Dhaka's annual average in 2024 was on the order of 78 micrograms per cubic metre — roughly fifteen times the guideline — and during the dry winter weeks of early 2025 the city repeatedly ranked first in the world for air pollution on daily monitoring indices (IQAir, 2025; The Daily Star, 2025). A resident of Dhaka is therefore not exposed to a marginal exceedance of a health threshold but to a chronic, order-of-magnitude one. Coarser particles matter too: poorly managed construction and demolition generate PM₁₀ and crystalline silica dust, the latter carrying its own well-documented long-term respiratory risk for those repeatedly exposed, workers on and residents beside the sites most of all.

3. Construction inside the source apportionment

The claim that construction belongs among the leading sources is not an inference from its visibility; it is what source-apportionment studies of Dhaka's particulate load report. Analyses of the city's PM_{2.5} consistently list construction activity and road dust alongside brick kilns, vehicular exhaust, and the burning of household solid fuels as principal contributors (Ecohubmap, n.d.; IQAir, 2025). Construction is not a marginal entry on that list. It is structural to a city in the middle of a building boom, and its contribution is continuous rather than confined to a single industry's season.

Two features of Dhaka's situation amplify the construction signal beyond what the raw activity would suggest. The first is geological. Dhaka sits on river-delta sand and fine sediment — material that is easily lifted, easily broken, and easily resuspended whenever ground is

disturbed. Every excavation, every load of aggregate, every truck dragging mud onto a road becomes a reservoir of particles that traffic and wind then return to the air again and again. The second is meteorological. Construction intensifies in the cool, dry winter, which is exactly the season when low wind speeds and temperature inversions trap pollutants near the ground and prevent them from dispersing. The result is a seasonal coincidence that makes the dry-season air far worse than the sum of its sources would imply: more dust is generated at precisely the time the atmosphere is least able to clear it. Large public infrastructure works, including the metro rail and the elevated expressway, have been flagged as significant ongoing dust sources in their own right, concentrating the exposure along the corridors where the most people pass.

4. Why a first-order source stays uncounted

If construction is this large a contributor, its low profile in public debate demands an explanation. Part of the answer is perceptual. Kiln smoke reads unambiguously as pollution — it is a plume, it comes from a chimney, it belongs to an identifiable industry that can be named and blamed. Construction dust reads as mess. It has no single chimney and no single owner; it is diffuse, produced by thousands of sites and the traffic around them, and it is bound up with development, which the same public broadly wants. A source that is everyone's and no one's, and that is a by-product of progress rather than of a disfavoured trade, is easy to treat as the unavoidable cost of a growing city rather than as a regulable harm.

Part of the answer is regulatory. Dust control at the level of the individual site — screening, covering loads, wetting down, washing wheels, hoarding demolition — is cheap, well understood, and routine in cities that enforce it. Its absence in Dhaka is not a technical failure but an accountability one: the harm is externalised onto the street and the lungs of passers-by, while the cost of preventing it would fall on the builder. As long as the dust is seen as nuisance rather than exposure, and as long as no one is required to internalise its cost, the rational move for each site is to do nothing, and the aggregate is the air Dhaka breathes. Making construction dust count — in monitoring, in attribution, and in enforceable site rules — is therefore less a scientific problem, which is largely solved, than a political and perceptual one.

5. Making the invisible visible: the case for the package

This diagnosis is what determined the form of the coursework project the essay accompanies. If the obstacle to treating construction as pollution is that people do not see it as pollution, then the useful intervention is not another abstract account of PM2.5 but a piece of work that returns the viewer's attention to a harm they have stopped noticing. A news-style video package was the right instrument for that. Rather than narrating the problem in the abstract, the package takes the viewer to where construction meets the street — the broken roads, the uncovered trucks, the haze at eye level — pairs those images with an expert able to explain what the particles are and how they enter the body, and keeps the runtime short enough to hold a general audience. The aim was journalism rather than lecture: show the thing, let an authority interpret it, and leave the viewer understanding a source of harm they walk through daily.

The expert interview carried the analytic weight, because a package like this lives or dies on whether a credible authority will put the science on record in plain language; arranging and shaping that interview so its answers would cut cleanly into a short format was the journalistic core of the work. The production discipline a tight package demands — deciding what to cut, how long to hold a shot, where the expert's words should land — is the same editorial judgment that long-form print reporting requires, applied to a different medium. The essay and the package are two treatments of one argument: that construction is a real and preventable source of Dhaka's dangerous air, and that the first step to controlling it is to make people count it as pollution at all.

6. Conclusion

Dhaka's air crisis is usually told as a story about kilns and traffic. That story is not wrong, but it is incomplete in a way that matters, because it leaves out a source that is structural, continuous, seasonally amplified by the delta's own geology and weather, and — crucially — preventable with measures that are already standard elsewhere. Construction dust stays uncounted not because the science is unsettled but because the harm is diffuse, ownerless, and bound up with development the public wants. Changing that begins with perception: with counting construction among the named sources, attributing its share honestly, and requiring the sites that produce the dust to bear the cost of controlling it. The package this essay accompanies was an attempt at the first move — to make a form of pollution people breathe every day visible enough to argue about.

References

- Ecohubmap. (n.d.). Air pollution in Dhaka, Bangladesh: Causes, consequences, and solutions. <https://www.ecohubmap.com/hot-spot/air-pollution-in-dhaka-bangladesh>
- IQAir. (2025). Dhaka air quality and PM2.5 reports [Newsroom analyses, 2025]. <https://www.iqair.com/>
- The Daily Star. (2025, January). Dhaka's deadly air: What we know and what we can do. The Daily Star. <https://www.thedailystar.net/opinion/focus/news/what-we-know-and-what-we-can-do-3850041>
- World Health Organization. (2021). WHO global air quality guidelines: Particulate matter (PM2.5 and PM10), ozone, nitrogen dioxide, sulfur dioxide and carbon monoxide. World Health Organization.
- Hossain, M. M. (Prod.). (2025). Building a crisis: Construction-driven air pollution in Dhaka [Video package]. NSC2283 Environmental Science group project ("Climate Catalysts"), University of Liberal Arts Bangladesh. <https://youtu.be/PvkRLinhKmk>